MIL-D-28728B
3 September 1986
SUPERSEDING
MIL-D-28728A
22 September 1972

MILITARY SPECIFICATION

DIAL, CONTROL, MULTI-TURN COUNTERS GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for manually operated, illuminated and nonilluminated, brake and nonbrake, multi-turn counters control dials (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

A-A-1830 - Tape, Pressure-Sensitive Adhesive, Box Closure
C-F-206 - Felt Sheet, Cloth, Felt, Wool, Pressed
P-C-444 - Cleaning Compound, Solvent Soluble, Grease Emulsifying
TT-T-266 - Thinner, Dope and Lacquer (Cellulose-Nitrate)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX 8112), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5355

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MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-P-116 - Preservation, Methods of.
MIL-L-3661 - Lampholders, Indicator Lights, Indicator Light Housings, and Indicator Light Lenses, Lens Style LC46.
MIL-L-6081 - Lubricating Oil, Jet Engine.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts); Packaging of.
MIL-C-25050 - Colors, Aeronautical Lights and Lighting Equipment, General Requirements for.
MIL-L-27160 - Lighting, Instrument, Integral, White.
MIL-C-45662 - Calibration System Requirements.
MIL-H-83282 - Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft Note Code Number H-537.

(See supplement 1 for list of associated specification sheets.)

STANDARDS

FEDERAL

FED-STD-3 - Colors, Aeronautical Lighting.
FED-STD-H28/2 - Screw-Thread Standards for Federal Services, Section 2, Unified Inch Screw-Threads UN and UNR Thread Forms.
FED-STD-H28/20 - Screw-Thread Standards for Federal Services, Section 20, Inspection Methods for Acceptability of UN, UNR, UNJ, M and MJ Screw-Threads.
FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-411 - Aircrew Station Signals.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MS51021 - Setscrew-Hexagon Socket, Cup Point, Corrosion-Resistant Steel, Passivated, UNC-3A, Plain and Self-Locking

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM

ASTM G23 - Operating Light-Exposure Apparatus (Carbon-Arc Type) With and Without Water for Exposure of Nonmetallic Materials.

ASTM D822 - Operating Light and Water Exposure Apparatus (Carbon-Arc Type) for Testing Paint, Varnish, Lacquer and Related Products, Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references other than specification sheets cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern. If specific requirements specified herein are not required for any dial, it shall be so indicated on the specification sheet.

3.2. First article. When specified, a sample shall be subjected to first article inspection (see 4.5 and 6.4).

3.3 Materials. Materials shall be as specified (see 3.1). However, when a definite material is not specified, a material shall be used which will enable the dials to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Plastic. Unless otherwise specified (see 3.1), plastic shall be in accordance with MIL-M-14.

3.3.2 Metals. All metal parts shall be of a corrosion-resistant material, or shall be suitably treated to resist corrosion.

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3.4 Design and construction.

3.4.1 Form and dimensions. Dials shall be made in the form and within the dimensions specified (see 3.1). If the cap has a configuration which might be construed as a pointer or indicator, any orientation mark, such as an arrow or dot, shall be in line with such configuration.

3.4.1.1 Concentricity and perpendicularity. Unless otherwise specified (see 3.1), the periphery of the dial shall be concentric to the rotational axis within 0.025 inch full indicator movement (FIM). The back plane of the dial shall be perpendicular within $\pm$ 1 degree to the rotational axis of the dial.

3.4.1.2 Threads. All threads shall be right hand and in accordance with FED-STD-H28/2. Acceptability of screw threads shall be in accordance with FED-STD-H28/20, system (22).

3.4.2 Insert (when applicable). The insert is applicable when the shaft bore is fabricated from material other than metal. The insert shall be non-rotatable, metallic, full cylinder, completely forming the shaft hole. At the option of the supplier, the insert may be molded in or pressed in the center of the dial body. It shall be provided with two UNC-3B threaded holes for set-screws as specified in 3.4.3. A chamfer of 1/32 $\pm$ 1/64 inch shall be provided in the insert end which admits the shaft.

3.4.3 Setscrews. Unless otherwise specified (see 3.1), locking type, cup point setscrews shall be provided within all dials. The setscrews shall conform to MS51021 for corrosion-resistant steel (CRES). Self-locking insert shall be in accordance with MIL-F-18240.

3.4.4 Color.

3.4.4.1 Nonilluminated dials. The color of nonilluminated dials shall be as specified (see 3.1), in accordance with FED-STD-595, and shall remain within the specified limits throughout the tests specified herein. The color for plastic dials shall not be applied by coating or painting, but shall be incorporated in the material.

3.4.4.2 Illuminated dials. The lens color shall be as specified in the specification sheets and shall be evenly distributed over the lens face when lighted or unlighted. The lens color shall be in accordance with the following specifications:

3.4.4.2.1 Level I colors.

COLORS	2200° KELVIN
RED	Type I (Aviation) in accordance with MIL-C-25050
YELLOW	Type I (Aviation) in accordance with MIL-C-25050
GREEN	Type I (Aviation) in accordance with MIL-C-25050
BLUE	Category I (Aviation) in accordance with FED-STD-3
WHITE	White in accordance with MIL-L-27160

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3.4.4.2.2 Level II colors.

COLORS	2200° KELVIN
RED	Type II (identification) in accordance with MIL-C-25050
YELLOW	Same as Level I
GREEN	Type II (identification) in accordance with MIL-C-25050
BLUE	Symbol B in accordance with MIL-L-3661
WHITE	Category I (Aviation) in accordance with FED-STD-3

3.4.5 Lamp circuitry (when applicable). The lamp circuit(s) shall be independent and shall be insulated from the assembly housing and dial case.

3.4.5.1 Lamp replacement. The lamps shall be replaceable from the front of the panel, with the total assembly or housing mounted to the panel. No special tools shall be required to replace the lamp.

3.4.5.2 Lamp retention. The lamps shall be held captive within the removable portion of the light assembly.

3.4.5.3 Lamp contact and springs. Lamp contacts shall be of a copper base alloy and shall be suitably plated. Lamp contact springs shall be beryllium copper suitably plated or shall be corrosion-resistant steel. When the contact and contact spring are combined as an integral part (one piece of metal), copper beryllium alloy, suitably plated, shall be used. Lamp contact(s) shall be of the multiple contact design; that is serrated or otherwise designed to provide for more than one bearing on the mating tip of the applicable lamp when installed. Single spike, flat button contacts shall not be used.

3.4.5.4 Lamp type. Unless otherwise specified (see 3.1), light assembly portion shall accept type T-1-3/4 or T-1 midget flange base incandescent lamps.

3.4.6 Weight. The weight shall be as specified (see 3.1).

3.4.7 Method of count. The method of count shall be concentric scale or digital dial, as specified (see 3.1).

3.4.8 Luminance. When specified (see 3.1), the measured photometric brightness in foot lamberts shall be in accordance with MIL-STD-411 (see 4.7.2).

3.4.9 Readout accuracy. When dials are tested as specified in 4.7.3, the readout accuracy shall be as specified (see 3.1).

3.5 Torque.

3.5.1 Insert (when applicable). When dials are tested as specified in 4.7.4.1, there shall be no movement of the insert, cracking, or other failure.

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3.5.2 Setscrew. When dials are tested as specified 4.7.4.2, there shall be no failure of the thread in the insert of the dial, setscrew threads, or setscrew socket heads. Insert or setscrew threads and sockets that are stripped, broken, or deformed shall be considered as failure of the thread.

3.5.3 Operational. When tested as specified in 4.7.4.3 the movement of the dial shall be smooth and unencumbered within the specified torque of 3.1.

3.5.4 Locking brake (when applicable). With the brake locked, test as specified in 4.7.4.3. There shall be no slippage or movement of the dial, or damage to the locking mechanism.

3.6 Environmental resistance. All dials shall withstand environmental tests as specified in 4.7.5 without failures, and degradation of torque and accuracy requirements. There shall be no cracking, chipping, spalling (see 6.3.3), crazing (see 6.3.4), peeling, illegibility of marking, change in color or luster beyond the specified limits, or destructive corrosion (see 6.3.5), after dials have been subjected to the following tests:

TESTS	TEST PARAGRAPHS
Solvents and fluids	4.7.5.1
Moisture resistance	4.7.5.2
Salt spray (corrosion)	4.7.5.7

3.6.1 Nonmetallic coating adhesion. Nonmetallic coated component parts of the dials, when tested (see 4.7.5.3) shall show no more than 25 percent loss of the coating.

3.6.2 Vibration. When dials are tested in accordance with 4.7.5.4, there shall be no degradation of component parts and the unit shall operate without binding.

3.6.3 Shock. When dials are tested in accordance with 4.7.5.5, there shall be no damage to component parts and the unit shall operate without binding.

3.6.4 Thermal shock. Dials shall be temperature conditioned in accordance with 4.7.5.6 without failures of component parts.

3.6.5 Dust. Dials shall be exposed to desert conditions in 4.7.5.8 without failures. The unit shall operate without binding.

3.7 Life cycle (rotational). Dials shall be mechanically operable after 2,000,000 revolutions of the knob (see 4.7.6).

3.8 Flammability. When dials are tested as specified in 4.7.7, they shall not support visible flame or combustion for more than 30 seconds after removal of the flame.

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3.9 Marking. Unless otherwise specified (see 3.1), dials shall be permanently and legibly marked with the military part number and manufacturer's source code. The markings may be sunken or raised characters stamped in contrasting color ink and shall be placed on a flat surface in a position not noticeable to the operator under normal operating conditions. When space does not permit the marking as specified above, the marking shall be applied or stamped on tape or label and the tape or label affixed to the dial. Paper labels shall not be used. Other markings which in any way interfere with, obscure, or confuse those specified herein are prohibited. The marking shall remain legible after all tests.

3.10 Workmanship. Dials shall be processed in such a manner as to be uniform in quality and shall be free from pits, cracks, chips, and other defects that will affect life, serviceability, appearance, or any combination of these.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. The manufacturer shall insure that test and inspection facilities of sufficient accuracy, quality and quantity are established and maintained to permit performance of required inspections.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Materials inspection (see 4.3).
b. First article inspection (see 4.5).
c. Quality conformance inspection (see 4.6).
d. Inspection of packaging (see 4.6.2).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials used in fabricating the dials are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202 and MIL-STD-810.

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4.5 First article inspection. Unless otherwise specified (see 6.2), six (6) units shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in Table I in the order shown.

TABLE I. FIRST ARTICLE INSPECTION

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH	NUMBER OF SAMPLE UNITS TO BE INSPECTED
Visual and mechanical examination	3.1, 3.4 to 3.4.7 incl, 3.9 and 3.10	4.7.1	1 thru 6
Luminance	3.4.8	4.7.2	3 and 4
Readout accuracy	3.4.9	4.7.3	3 and 4
Torque:	3.5	4.7.4	—
Insert (when applicable)	3.5.1	4.7.4.1	1 and 2
Setscrew	3.5.2	4.7.4.2	1 and 2
Operational	3.5.3	4.7.4.3	1, 2 and 6
Locking brake	3.5.4	4.7.4.3	1, 2 and 6
Environmental resistance	3.6	4.7.5	—
Solvent and fluids	3.6	4.7.5.1	3 and 4
Moisture resistance	3.6	4.7.5.2	1 and 2
Coating adhesion	3.6.1	4.7.5.3	1 and 2
Vibration	3.6.2	4.7.5.4	2, 3 and 4
Shock	3.6.3	4.7.5.5	3 and 4
Thermal Shock	3.6.4	4.7.5.6	3 and 4
Salt spray	3.6	4.7.5.7	3 and 4
Dust	3.6.5	4.7.5.8	6
Life cycle	3.7	4.7.6	1 and 5
Flammability	3.8	4.7.7	1

4.5.3 Failures. One or more failures shall be cause for refusal to grant first article approval; however, lamp electrical failure does not constitute a failure.

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A and B inspections.

4.6.1.1 Inspection lot. An inspection lot shall consist of all dials of the same part number and functional type produced under essentially the same conditions, and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examination specified in Table II.

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TABLE II. GROUP A INSPECTION

TEST OR INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH	AQL (PERCENT DEFECTIVE) MAJOR	MINOR
Visual and mechanical examination	3.1, 3.4 to 3.4.8 incl, 3.9 and 3.10	4.7.1, 4.7.2	1.0	4.0

4.6.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in Table II. Minor defects shall be as defined in MIL-STD-105. Major defects shall be design, dimensions or form (see 4.7.1).

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.6.1.3 Group B inspection. Group B inspection shall consist of the examinations and tests specified in Table III, in the order shown, and the sample shall be selected from inspection lots that have passed the group A inspection.

TABLE III. GROUP B INSPECTION

TEST OR INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
Torque:		
Insert (when applicable)	3.5.1	4.7.4.1
Brake (when applicable)	3.5.4	4.7.4.3
Readout accuracy	3.4.9	4.7.3

4.6.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S-2. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The AQL shall be 6.5 percent defective.

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4.6.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.6.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection may be delivered on the contract or purchase order, if the lot is accepted.

4.6.2 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of MIL-E-17555.

4.7 Methods of inspection.

4.7.1 Visual and mechanical examination. Dials shall be examined to verify that the design, form, dimensions, color, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.4 to 3.4.8 inclusive, 3.9 and 3.10.).

4.7.2 Luminance. An average of three readings shall be made across the face of the lens when unit is illuminated for compliance with the requirements specified in 3.4.8.

4.7.3 Readout accuracy. Turn counting dials shall be mounted by their normal mounting means on a suitable fixture to measure the shaft rotation to an accuracy of $\pm 0.1^{\circ}$. The dial scale and the shaft in relation to the fixture shall be set at 0° as starting point. Rotate the knob to the full count of the dial and note the reading on the fixture. Rotate the knob in the opposite direction to the starting point of the dial and the fixture stopping at intermediate steps of the turn dial, noting the shaft in the relation to the dial reading (see 3.4.9).

4.7.4 Torque.

4.7.4.1 Insert. The dials shall be mounted on a corrosion-resistant shaft and held in a suitable jig. A torque of 15 inch-pounds for 1/8 inch shaft hole and for all No. 4-40 UNC-3A setscrews, and 30 inch-pounds for 1/4 inch shaft hole shall be applied to the shaft in a clockwise direction, then in a counterclockwise direction for a period of 1 minute in each direction. The dials shall be examined for evidence of movement of the insert within the dial, cracking or other failure (see 3.5.1).

4.7.4.2 Setscrew. A suitable wrench shall be inserted into the socket head of the setscrew of the dial. A torque of 3 inch-pounds shall be applied for No. 4-40 UNC-3A setscrews, 10 inch-pounds for No. 6-32 UNC-3A setscrews and 15 inch-pounds for No. 8-32 UNC-3A setscrews. Examine the threads and socket for failures (see 3.5.2).

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4.7.4.3 Operational and locking brake. With the dial counter mounted on suitable panel, without shaft engagement, rotate the knob with a suitable wrench to measure the rotational torque of the knob (see 3.5.3 and 3.5.4).

4.7.5 Environmental resistance. Dials shall be mounted by their normal mounting means on a corrosion-resistant shaft of the largest nominal diameter that will fit the shaft hole of the dial during the tests specified in 4.7.5.1 through 4.7.5.7 (see 3.6).

4.7.5.1 Solvents and fluids. The following materials shall be used:

a. Lacquer Thinner, TT-T-266.
b. Engine Oil (Grade 1010), MIL-L-6081.
c. Grease Cleaning Compound, P-C-444.
d. Hydraulic Fluid, MIL-H-83282.
e. Felt Pad, C-F-206, Classification 12R1 or 12R2.

4.7.5.1.1 Procedure. A felt pad 1/4 inch thick shall be saturated with the fluid. The felt pad shall then be placed on the face of the knob and held down by a uniformly distributed force of 1.5 pounds for one minute. At the termination of this time, the felt pad shall be removed and a clean, lint-free, dry cloth shall be used to wipe the face of the knob and examined (see 3.6).

4.7.5.2 Moisture resistance. Dials shall be tested in accordance with Method 106 of MIL-STD-202. The following details shall apply:

a. Initial measurements - Not applicable.
b. Loading voltage - Not applicable.
c. Final examination - Dials shall be examined for visible deterioration.

4.7.5.3 Nonmetallic coating adhesion. After completion of moisture resistance test as specified in 4.7.5.2, the dials shall be wiped dry and within 1 minute several lines shall be scratched on nonmetallic coated parts to a depth of at least 0.003 inch with a sharp pointed stylus. Two parallel lines at least 1/4 inch long and 1/8 inch apart and two other equal parallel lines intersecting the first two at right angles, shall be scratched on several smooth surfaces (cap, and dial plate, when available). A single line shall also be scratched on the crest of two or more grip serrations, parallel to the grooves. A 1/2 inch wide strip of tape conforming to A-A-1830 of sufficient length to cover the lines shall be firmly pressed with the adhesive side against the scratched lines. The tape shall then be abruptly pulled off in one motion. The surface of the dials shall then be examined (see 3.6).

4.7.5.4 Vibration. Dials shall be tested in accordance with Method 204 of MIL-STD-202. During this test, the brake lever shall be in a locked position. The following details shall apply:

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a. Method of mounting - Specimens shall be rigidly mounted on the horizontal platform of a vibration machine in such a manner that the applied vibration shall be as follows:

1. One-third of the units (to the nearest integral number) shall have the direction of vibration parallel to the pin length.
2. The same numbers of units shall have the direction of vibration perpendicular to the largest surface.
3. The remainder of the sample units shall have the direction of vibration perpendicular to the pin length and parallel to the largest surface.

b. Test condition - B, unless otherwise specified (see 3.1).

4.7.5.4.1 Examination after test. Dials shall be manually operated and examined for binding, backlash readout tolerance, wear and cracks of component parts.

4.7.5.5 Shock. Dials shall be tested in accordance with 4.7.5.5.1, unless otherwise specified (see 4.7.5.5.2). During these tests, the specimen shall be suitably mounted on the test platform and the brake lever shall be in a locked position.

4.7.5.5.1 Shock (specified pulse). Dials shall be tested in accordance with Method 213 of MIL-STD-202. The following details shall apply:

a. Test condition letter - I, unless otherwise specified (see 3.1).
b. Measurements before and after test - Same as 4.7.5.4.1.

4.7.5.5.2 High-impact shock (when specified). Dials shall be tested in accordance with Method 207 of MIL-STD-202. Examine the dials after test in accordance with 4.7.5.4.1.

4.7.5.6 Thermal shock. Dials shall be tested in accordance with Method 107 of MIL-STD-202, except in step 1 the temperature shall be -62°C. The following details shall apply:

a. Test conditions letter - A.
b. Measurements before and after cycling - Same as 4.7.5.4.1.

4.7.5.7 Salt spray (corrosion). Dials shall be tested in accordance with Method 101 of MIL-STD-202. The following details shall apply:

a. Applicable salt solution - 20 percent solution.
b. Test condition - B.
c. Examination after exposure - Dials shall be examined for visible corrosion.

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4.7.5.8 Dust. Dials shall be tested in accordance with Method 510.1, Procedure I of MIL-STD-810. Once every hour during test steps 1, 2 and 3 rotate the knob 10 revolutions forward and return. Examine the test specimen in accordance with 4.7.5.4.1.

4.7.6 Life cycle (rotational). This test shall be conducted in room temperature. With dials mounted on a suitable panel and no load on shaft, rotate the knob forward the number of turns specified in Method of count (see 3.1) and return at speeds not exceeding maximum operational RPM. Total number of cycles shall be as specified (see 3.1).

4.7.7 Flammability. The dials shall be tested in accordance with Method 111 of MIL-STD-202. The dial shall be mounted on a horizontal shaft located on the center of a 1 foot square, vertically mounted panel, with the back surface of the dial not less than 1/8 nor more than 1/4 inch from the surface of the panel. The axis of the flame shall be in the plane of the axis of the shaft and at an angle of 45 degrees below the axis of the shaft. The flame shall be applied to the lowest portion of the dial at the junction of the dial body with the cap or top surface of the dial (see 3.8).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. Multi-turn dials conforming to this specification are intended for use with electrical, electronics, communications and allied equipment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Title, number and date of the applicable specification sheet and the complete military part number (see 3.1).
c. Number of first article samples to be submitted if other than specified in 4.5.
d. Levels of preservation, packaging, packing and marking (see section 5).

6.3 Definitions. The following definitions are applicable to this specification:

6.3.1 Cycle. Number of revolutions of the knob to full count on the turns dial scale and return.

6.3.2 Gear ratio. Number of knob turns to one turn of the shaft.

6.3.3 Spalling. Splits or splinters on the surface of the dial.

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6.3.4 Crazing. Minute cracks on the surface of the dial.

6.3.5 Destructive corrosion. And any type of corrosion which prevents or hinders free mounting or dismounting and normal operation of the dial control.

6.4 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.5 Subject term (keyword) listing.

Control Dial
Levels of Color
Operating Light-Exposure
Lamp Retention
Multiturn Counters
Solvents and Fluids

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of changes.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - IS

Review activities:
Army - AV, MI
Navy - AS, OS, SH
Air Force - 80, 82

(Project 5355-0188)

User activities:
Army - AR, ME
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-28728B	DIAL, CONTROL, MULTI-TURN COUNTER GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTICE OF GROUP VALIDATION

INCH-POUND

MIL-D-28728B
NOTICE 1
25 February 1991

MILITARY SPECIFICATION

DIAL, CONTROL, MULTI-TURN COUNTERS GENERAL SPECIFICATION FOR

The following military specification sheets of MIL-D-28728B have been reviewed and are determined to be valid for use in acquisition as listed below:

Document	Date
MIL-D-28728/2B	03 September 1986
/3A	06 December 1971
/7	15 May 1970

Preparing activity:
Navy - EC

Agent:
DLA-GS

AMSC N/A

FSC 5355

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